# BirthVerse - Technical & User Documentation

System Architecture, Algorithms, API Reference & User Guide

**Developed by Zabiada Arman | October 2026**

System Name: BirthVerse - Historical & Temporal Life Analytics

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Version: 1.0.0

Language & Runtime: Python 3.10+ / HTML5 / CSS3 / ES6 JavaScript

## 1. System Overview

BirthVerse is a web-based temporal analytics and historical exploration system written in Python. When a user submits their birth date (and optional name), the system computes an extensive profile across six analytical dimensions:

1. Exact Chronological & Cumulative Age (Years, Months, Weeks, Days, Hours, Minutes, and Live Seconds).

2. Historical Events & Notable Shared Birthdays occurring on that calendar date.

3. Birth Year Cultural & Demographic Time Capsule (Global population growth, technology, music, cinema, and economic benchmarks).

4. Physiological & Astrophysical Telemetry (Heartbeats, respiratory cycles, sleep cycles, solar orbit distance, and galactic travel distance).

5. Astronomical Birth Chart & Planetary Ages (Synodic Moon phase, illumination percentage, zodiac profiles, birthstone, and orbital ages on all eight planets).

6. Landmark Temporal Milestones (Exact dates for 10,000 days alive, 100,000 hours, 1 billion seconds, and an exportable Life Summary Card).

## 2. Project Directory Structure

```
birthday/
|---- app.py                                  # Main Python HTTP server & REST API router
|---- birthday_engine.py                      # Temporal, astronomical & biometric computa
|---- history_data.py                         # Hybrid Wikimedia API client & curated hist
|---- README.md                               # Quick-start guide & project overview
|---- templates/
|   `-- index.html                          # Main application layout & semantic structu
|---- static/
|   |---- css/
|   |   `-- styles.css                      # Architectural dark theme, layout grids & p
|   `-- js/
|       `-- app.js                          # Client-side API controller, 1 Hz live tick
`-- docs/
    |---- BirthVerse_Project_Proposal.docx    # Project Proposal (Microsoft Word)
    |---- BirthVerse_Project_Proposal.pdf     # Project Proposal (PDF)
    |---- BirthVerse_Project_Documentation.docx   # Technical Documentation (Microsoft Word)
    `-- BirthVerse_Project_Documentation.pdf    # Technical Documentation (PDF)
```

## 3. Backend Architecture & Module Specification

### 3.1 Web Server & API Router (app.py)

app.py implements ChronosHTTPRequestHandler (subclassing http.server.SimpleHTTPRequestHandler) to provide a zero-dependency web server capable of serving both static assets and dynamic JSON API endpoints:

- GET /: Serves the main application interface (templates/index.html).
- GET /api/analyze?date=YYYY-MM-DD&name= OptionalName: Validates query parameters, invokes compute_birthday_profile(), and returns a structured JSON response.
- POST /api/analyze: Accepts a JSON request body {"date": "YYYY-MM-DD", "name": "..."} and returns the computed profile.
- GET /static/*: Safely serves stylesheets and JavaScript files with strict path-traversal protection and appropriate MIME headers.

### 3.2 Temporal & Astronomical Computation Engine (birthday_engine.py)

birthday_engine.py contains the core mathematical functions:

1. Exact Calendar Age Decomposition:

Computes exact calendar years, months, and days by accounting for variable month lengths (calendar.monthrange) and leap years (calendar.isleap), alongside total elapsed seconds (elapsed_seconds = t_now - t_birth).

2. Synodic Lunar Phase Algorithm (calculate_moon_phase):

Uses the astronomical synodic month constant (S = 29.530588853 days) anchored to the reference New Moon epoch of January 6, 2000 at 18:14 UTC:

- Cycle Position (phi) = ((t_birth - t_epoch) mod S) / S (ranging from 0.0 to 1.0)
- Illumination (%) = ((1 - cos(2 * pi * phi)) / 2) * 100

3. Planetary Orbital Age Calculation:

Divides total Earth days lived (D_lived) by each planet's heliocentric orbital period P_planet (Mercury: 87.969 d, Venus: 224.701 d, Earth: 365.256 d, Mars: 686.980 d, Jupiter: 4332.589 d, Saturn: 10759.22 d, Uranus: 30688.5 d, Neptune: 60182.0 d) and projects the exact calendar date of the user's next planetary birthday.

4. Physiological & Cosmic Velocity Models:

- Heartbeats: elapsed_seconds * (72 / 60)
- Respiratory cycles: elapsed_seconds * (16 / 60)
- Solar Orbit Distance: elapsed_seconds * 29.78 km/s
- Milky Way Galactic Drift: elapsed_seconds * 230.0 km/s

5. Temporal Milestone Projection:

Adds fixed datetime.timedelta offsets (1,000 days, 5,000 days, 10,000 days, 100,000 hours, 250,000 hours, 500,000 hours, 1,000,000,000 seconds, 2,000,000,000 seconds) to the birth timestamp to determine whether each milestone has been completed or how many days remain until it occurs.

### 3.3 Historical & Demographic Archive (history_data.py)

history_data.py implements a resilient two-tier data architecture:

- Tier 1 (Live Wikimedia Feed): Queries https://api.wikimedia.org/feed/v1/wikipedia/en/onthisday/all/{MM}/{DD} with a 2.5-second timeout to fetch live historical events and notable births.
- Tier 2 (Embedded Curated Archive): Provides curated exact-date highlights, monthly thematic events across World History, Science & Space, and Culture, plus decade-by-decade demographic and economic time capsules (1940s-2020s) with linear

population interpolation. This guarantees instantaneous, error-free operation even in offline environments.

## 4. Frontend Design & Interactive Engine

### 4.1 Architectural Dark Interface (static/css/styles.css)

- Visual Hierarchy: Built on a deep obsidian-indigo palette (#050711) with translucent glassmorphic surfaces (backdrop-filter: blur(18px)), subtle 1px specular borders, and tabular monospace numerals (font-variant-numeric: tabular-nums) so real-time ticking numbers never jitter horizontally.
- Single-Line Navigation Dock: A sticky floating navigation bar with 0.45s cubic-bezier(0.22, 1, 0.36, 1) hover transitions allows effortless jumping between analytical sections.
- Print Stylesheet (@media print): Automatically hides navigation controls and background canvases when printing or saving the Official Life Summary Card as a PDF.

### 4.2 Client-Side Controller (static/js/app.js)

- 1 Hz Real-Time Ticker (startLiveCounters): Recalculates elapsed seconds, minutes, hours, days, heartbeats, breaths, blinks, solar orbit distance, and galactic distance every 1,000 ms relative to birth_timestamp_ms.
- Dynamic SVG Lunar Phase Renderer (renderMoonSVG): Constructs an accurate SVG path using elliptical arc terminators (A rx ry ...) corresponding to waxing, waning, crescent, quarter, gibbous, and full lunar phases.

## 5. REST API Reference

### Endpoint: GET /api/analyze

Query Parameters:

- date (required): Birth date in ISO format (YYYY-MM-DD).
- name (optional): User's display name.

Sample JSON Response (200 OK):

```
{
  "ok": true,
  "data": {
    "user": {
      "name": "Zabiada Arman",
      "birth_date_iso": "2000-10-28",
      "birth_date_formatted": "October 28, 2000",
      "birth_weekday": "Saturday",
      "serial_code": "BV-20001028-SC-9496"
    },
    "age_breakdown": {
      "years": 26,
      "months": 0,
      "days": 0,
      "total_days": 9496,
      "total_hours": 227924,
      "total_seconds": 820529200
    }
  }
}
```

# 6. Installation & User Guide

## 6.1 Running the Application Locally

1. Ensure Python 3.10+ is installed (python3 --version).

2. Navigate to the project directory:

   bash

   cd birthday

3. Launch the server:

   bash

   python3 app.py

4. Open http://127.0.0.1:8080 in any modern web browser.

## 6.2 Using the Interface

1. Enter Details: Type your name (optional) and select your birth date in the top control deck, then click Explore My Birthday.

2. Navigate Sections: Use the floating navigation bar to jump between *Overview*, *On This Day in History*, *Birth Year Capsule*, *Vital & Space Metrics*, *Moon, Zodiac & Planets*, *Milestones*, and *Summary Card*.

3. Filter Historical Events: Click the category pills (*All Events*, *World History*, *Science & Space*, *Culture & Arts*, *Notable Shared Birthdays*) to filter the historical chronicle.

4. Export or Print Summary: Scroll to the Official Birthday & Life Summary card at the bottom to copy a formatted text summary to your clipboard or print/save the card as a PDF.